SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1005	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	Yazhini D	**Reg. No.:**	192511413
Date:	10.10.26	**Duration:**	60 minutes

Code of Conduct

I _____Yazhini D / 192511413_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____Yazhini D_____

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. **Understand the Problem Statement**
   - Explain the problem in your own words.
   - Identify the objectives of the proposed system.
   - State the scope of the problem.
2. **Identify Key Stakeholders and Challenges**
   - Identify all the stakeholders involved in the system.
   - Explain the roles and responsibilities of each stakeholder.
   - Discuss the major challenges and issues faced by the stakeholders.
3. **Analyze the Current Approach**
   - Describe the existing system or current process.
   - Identify its strengths and limitations.
   - Analyze the problems or gaps that need to be addressed.
4. **Propose a Suitable Solution Using Software Engineering Principles**
   - Design a software-based solution for the given problem.
   - Describe the major functional modules of the proposed system.
   - Identify the functional and non-functional requirements.
   - Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.
5. **Justify the Chosen Methodology/Framework**

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
- Justify why the selected methodology/framework is suitable for the proposed system.
- Explain how it supports successful project development and maintenance.

6. **Discuss Expected Outcomes and Limitations**
   - Describe the expected benefits of the proposed solution.
   - Explain how the solution addresses the identified challenges.
   - Discuss potential risks, implementation challenges, and limitations.
   - Suggest possible future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework	Selects the most suitable SDLC/model/frame	Suitable methodology selected with	Methodology selected but	Outcomes, limitations, or future

Justification (10%)	work and provides strong technical justification.	adequate justification.	justification is weak.	enhancements are missing; report lacks organization and clarity.

Criteria	**Mark**
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25

Smart Personal Document Expiry and Renewal Management System

1. Understand the Problem Statement

Problem Explanation

People maintain documents such as passports, driving licences, insurance policies and certificates. Forgetting their expiry dates can cause penalties and inconvenience. This system stores document details, tracks expiry dates and sends reminders before renewal.

Objectives

- Register personal and organizational documents.
- Monitor expiry dates.
- Send advance reminders.
- Categorize documents by type and importance.
- Maintain renewal history and status.
- Store document information securely.
- Generate expiry reports.

Scope

The system helps individuals and organizations manage documents in one place. It provides reminders, renewal tracking and reports. Automatic renewal through external authorities is outside the initial scope.

2. Key Stakeholders and Challenges

- **Users:** Add documents and receive reminders.
- **Organization administrators:** Manage employee and organizational documents.
- **System administrators:** Maintain security, backups and user access.
- **Development team:** Develop, test and maintain the application.
- **Notification providers:** Deliver email or SMS reminders.
- **Document authorities:** Issue and renew official documents.

Challenges: Incorrect dates, missed reminders, data loss, privacy risks and managing many documents.

3. Analyze the Current Approach

Existing System

People use paper files, spreadsheets, calendars and phone reminders to track expiry dates.

Strengths

- Easy to use.
- Low cost.

- Suitable for a small number of documents.

Limitations

- Information is scattered across different locations.
- Manual tracking can cause errors.
- Important deadlines may be missed.
- Renewal history is difficult to maintain.
- Preparing reports requires extra effort.

Identified Gap: A centralized system is needed to manage document details, reminders, renewal history and reports automatically.

4. Proposed Software Solution

System Overview

The proposed web or mobile application stores document details, checks expiry dates and sends reminders. Users can view upcoming expiries, expired documents and renewal status through a dashboard.

Functional Modules

1. **User Management:** Registration and login.
2. **Document Management:** Add, edit and view documents.
3. **Category Management:** Group documents by type and importance.
4. **Expiry Monitoring:** Identify documents nearing expiry.
5. **Reminder Management:** Send configurable advance alerts.
6. **Renewal Tracking:** Record renewal status and history.
7. **Report Generation:** Display upcoming and expired documents.
8. **Secure Storage:** Protect document information and files.

Functional Requirements

- Register and update document details.
- Store issue and expiry dates.
- Generate reminders before expiry.
- Track renewal status and history.
- Generate expiry reports.
- Restrict access to authorized users.

Non-Functional Requirements

- **Security:** Protect personal information.
- **Usability:** Provide a simple interface.
- **Reliability:** Ensure consistent expiry checking.
- **Scalability:** Support more users and documents.
- **Maintainability:** Allow easy updates.
- **Performance:** Provide quick access to records.

Software Engineering Principles

- **Modularity:** Divide the system into separate modules.
- **Scalability:** Support future growth.
- **Maintainability:** Make updates easier.
- **Reliability:** Validate dates and track reminder failures.
- **Usability:** Provide simple navigation.
- **Security:** Use authentication and access control.

Basic Workflow

1. User logs in.
2. User registers a document.
3. System saves and validates the details.
4. System monitors the expiry date.
5. Reminders are sent before expiry.
6. User updates the renewal status.
7. Reports and renewal history are updated.

5. Justify the Chosen Methodology

Kanban Methodology

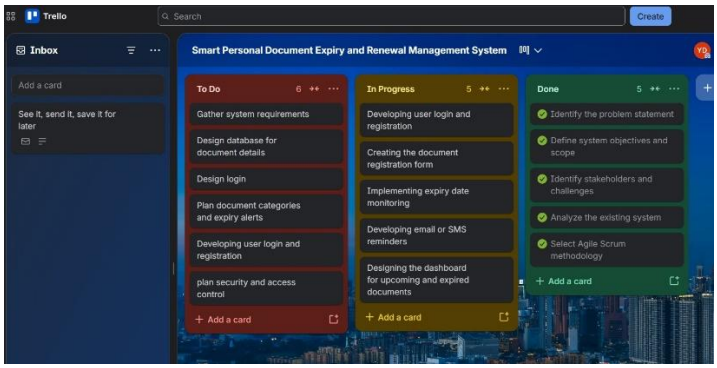
Kanban methodology is chosen for the Smart Personal Document Expiry and Renewal Management System because it helps organize tasks, track progress, and manage development efficiently. Trello is used to manage tasks through three lists: To Do, In Progress, and Done.

Implementation Plan

- To Do: Requirements gathering, database design, login design, and security planning.

- In Progress: User login, document registration, expiry monitoring, reminders, and dashboard development.
- Done: Problem identification, defining objectives and scope, stakeholder analysis, and existing system analysis.

Advantages

- Easy task management and progress tracking.
- Flexible workflow and prioritization.
- Early identification of problems.
- Improved productivity and teamwork.
- Continuous improvement throughout development.
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6. Expected Outcomes and Limitations

Expected Outcomes

- Fewer missed renewal deadlines.
- Better document organization.
- Improved compliance.
- Reduced manual work.
- Timely renewal of important documents.
- Centralized document management.

Limitations and Risks

- Incorrect dates may produce incorrect reminders.
- Email or SMS notifications may be delayed.
- Sensitive information requires strong security.
- Users must update renewal details.

- External authority integration may not be available.
- Maintenance and notification services may involve costs.

Future Enhancements

- Mobile application with push notifications.
- OCR to identify document details automatically.
- Calendar integration.
- Multilingual support.
- Integration with authorized renewal portals.
- Advanced reports and analytics.

Conclusion

The Smart Personal Document Expiry and Renewal Management System provides a centralized way to store document details, monitor expiry dates and send reminders. It reduces manual effort, improves organization and supports timely renewals. The Kanban methodology supports continuous workflow, visual task tracking, flexible prioritization and gradual improvement while maintaining security, reliability and usability.

References

1. Pascoe, C., Quinn, S., and Scarfone, K. (2024). The NIST Cybersecurity Framework (CSF) 2.0. National Institute of Standards and Technology. Read online.NIST
2. Atlassian. (n.d.). What is Kanban? A guide to the Kanban method. Atlassian Agile Coach. https://www.atlassian.com/agile/kanban
3. OWASP Foundation. (2025). Application Security Verification Standard (ASVS), Version 5.0.0. Official website.
4. National Institute of Standards and Technology. (2020). Security and Privacy Controls for Information Systems and Organizations (NIST SP 800-53, Revision 5). Read online. NIST
5. ISO/IEC/IEEE. (2024 review). ISO/IEC/IEEE 29148:2018 — Systems and Software Engineering: Requirements Engineering. Official standard page